

Genome-wide analysis of glucose-6-phosphate dehydrogenases in Arabidopsis

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Summary

In green tissues of plants under illumination, photosynthesis is the primary source of reduced nicotinamide adenine dinucleotide phosphate (NADPH), which is utilized in reductive reactions such as carbon fixation and nitrogen assimilation. In non-photosynthetic tissues or under non-photosynthetic conditions, the oxidative pentose phosphate pathway contributes to basic metabolism as one of the major sources of NADPH. The first and committed reaction is catalyzed by glucose-6-phosphate dehydrogenase (G6PDH). We characterized the six members of the G6PDH gene family in Arabidopsis. Transit peptide analysis predicted two cytosolic and four plastidic isoforms. Five of the six genes encode active G6PDHs. The recombinant isoforms showed differences in substrate requirements and sensitivities to feedback inhibition. Plastidic isoforms were redox sensitive. One cytosolic isoform was insensitive to redox changes, while the other was inactivated by oxidation. The respective genes had distinct expression patterns that did not correlate with the activity of the proteins, implying a regulatory mechanism beyond the control of mRNA abundance. Two cytosolic and one plastidic isoform were detected *in vivo* using zymograms, and the respective genes were identified using T-DNA insertion lines. The activity of a plastidic isoform was detected in all tissues including photosynthetic tissues despite its sensitivity to reduction observed *in vitro*. Genomic data, gene expression, and *in vivo* enzyme activity data were integrated with *in vitro* biochemical data to propose *in vivo* roles for individual G6PDH isoforms in Arabidopsis.

Keywords: glucose-6-phosphate dehydrogenase, oxidative pentose phosphate pathway, carbohydrate metabolism, isoenzyme, NADPH.

Introduction

In green tissues of plants during the light period, photosynthesis generates reduced nicotinamide adenine dinucleotide phosphate (NADPH), which is used in many reactions, for example, carbon fixation, fatty acid synthesis, and nitrogen assimilation. The oxidative pentose phosphate pathway (OPPP) is one of the major sources of NADPH in non-photosynthetic tissues as well as in photosynthetic tissues under darkness (Emes and Neuhaus, 1997; Thom *et al.*, 1998). The two dehydrogenases in the OPPP, glucose-6-phosphate dehydrogenase (G6PDH, EC 1.1.1.49) and 6-phosphogluconate dehydrogenase (6PGDH, EC 1.1.1.44), use NADP⁺ as a cofactor and generate two moles of NADPH during the conversion of one mole glucose-6-phosphate (G6P) to ribulose-5-phosphate. The reaction catalyzed by G6PDH is considered the regulatory step in the OPPP

because of its tight regulation by the redox balance, that is, NADPH/NADP⁺ (Johnson, 1972; Lendzian, 1980; Williams, 1980).

In plants, substantial evidence shows G6PDH to be involved in nitrogen assimilation (Bowsher *et al.*, 1992; Esposito *et al.*, 2001; Oji *et al.*, 1985; Wang *et al.*, 2003; Wright *et al.*, 1997), responses to stresses including oxidative stress (Debnam *et al.*, 2004; Hauschild and von Schaewen, 2003), pathogenesis (Batz *et al.*, 1998; Sindelar and Sindelarova, 2002; Tecsí *et al.*, 1994), salt stress (Nemoto and Sasakuma, 2000), and metal toxicity (Esposito *et al.*, 1998). Two subcellular localizations have been confirmed for plant G6PDHs, cytosolic and plastidic (Schnarrenberger *et al.*, 1973), and both isoforms are feedback inhibited by NADPH (Lendzian and Bassham, 1975; Schnarrenberger

et al., 1973). In chloroplasts, where photosynthesis generates NADPH, plastidic G6PDH isoforms are inactivated by thioredoxin reduction that involves two cysteine residues conserved only among plastidic isoforms (Fickenscher and Scheibe, 1986; Graeve *et al.*, 1994; Lenzian, 1980; Scheibe and Anderson, 1981; Wenderoth *et al.*, 1997; Wendt *et al.*, 2000). Thus for plastidic isoforms, two mechanistic principles for G6PDH regulation are known, feedback inhibition and reductive inactivation, providing the basis for an inverse relationship between photosynthesis and G6PDH activity. To date, only one study reports peroxisomal G6PDH (protein and activity) in pea leaves (*Pisum sativum*), but neither nucleotide nor amino acid sequences are available (Corpas *et al.*, 1998). The authors speculated that G6PDH as well as 6PGDH, the second enzyme in OPPP, are involved in the regeneration of NADPH required for the ascorbate-glutathione cycle occurring in the leaf peroxisome.

A G6PDH isoform not subjected to these regulatory mechanisms can be postulated and such an isoform could potentially play a role in providing NADPH in non-photosynthetic tissues or in green tissues independently of photosynthesis to support sudden increases in the demand for NADPH, for example, under oxidative stress (Esposito *et al.*, 2003; Wendt *et al.*, 2000). In their phylogenetic study, Wendt *et al.* (1999) showed that plastidic isoforms are separated into two groups, P1 and P2, which can be distinguished by specific antibodies. Distinct biochemical characteristics together with specific gene expression patterns were observed for a P1 and a P2 isoform in potato (Wendt *et al.*, 2000). Isoform P2 was less sensitive to both regulatory mechanisms by NADPH and the gene was ubiquitously expressed while the P1 gene was predominantly expressed in green tissues. The authors proposed that G6PDH isoforms belonging to the plastidic P1 group are tightly regulated by photosynthetic redox potential in green tissues, while those in the P2 group are less tightly regulated ubiquitous isoforms. A P2 isoform was also identified from barley roots with a low sensitivity to NADPH feedback inhibition, reinforcing this view (Esposito *et al.*, 2003).

Without a plant's genomic sequence, characterizing the full complement of isoforms for a given class of enzymes has not been possible in the past. In the post-genomic era, this limitation is no longer hampering genome-wide analysis. To identify all isoforms of G6PDH in a single plant ranging from those with major to the lowest activity and/or abundance, we took a genomic approach using Arabidopsis.

Results

Phylogenetic analysis

A previous phylogenetic analysis of G6PDH has been carried out by Wendt *et al.* (1999), which included three of the six Arabidopsis isoforms. In our analysis, we took advantage of

additional plant genome sequences and included all Arabidopsis and rice proteins translated from genes annotated as G6PDH and plant protein sequences that were deposited to the database since then (Figure 1a). The Arabidopsis

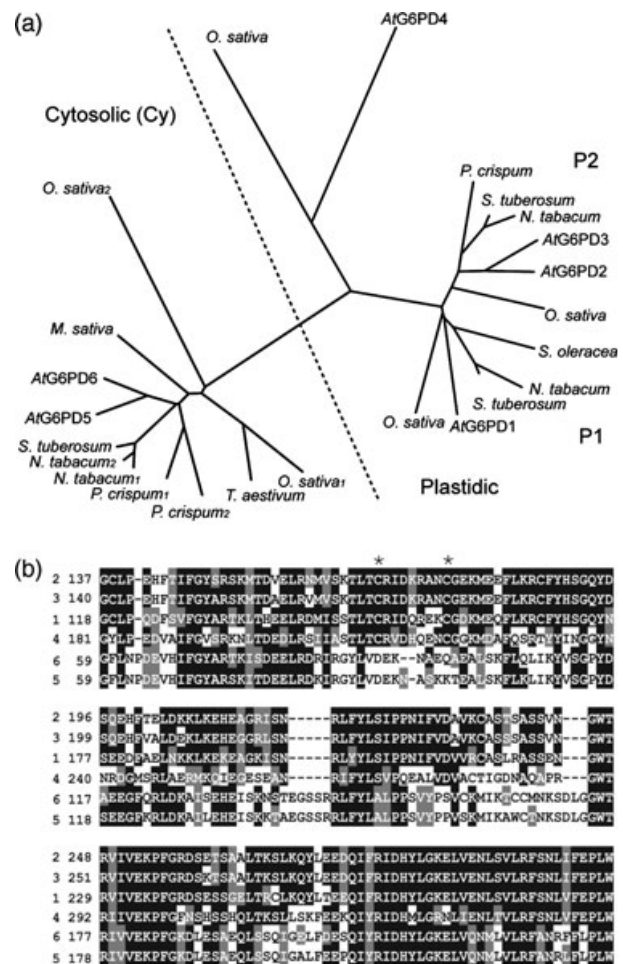


Figure 1. Plant G6PDH similarity.

(a) Plant G6PDHs. The mature protein sequences of G6PDHs were used. Abbreviations follow a previous report (Wendt *et al.*, 1999, Figure 3), Cy, cytosolic; P1/P2, plastidic isoforms. SwissProt accession numbers are provided below. The ID numbers of DNA loci on the TIGR website (The Institute for Genomic Research <http://www.tigr.org/tdb/e2k1/osa1/>) are given for the two rice isoforms and their amino acid sequences were translated from available genomic sequence; *Arabidopsis thaliana*: AtG6PD1, Q43727; AtG6PD2, Q9FY99; AtG6PD3, Q8L743; AtG6PD4, Q93ZW0; AtG6PD5, Q9LK23; AtG6PD6, Q9FJ15; *Medicago sativa*: AAB41552; *Nicotiana tabacum*: P1, CAA04994; P2, AAF87216; Cy¹, CAA04992; Cy², CAA04994; *Oryza sativa*: P1, AAS07054; P2, BAC84352; predicted plastidic, translated protein of locus 7230.t00022; Cy¹, translated protein of locus 4978.t0007; Cy², CAC09489, CAE02006, CAE03156; *Petroselinum crispum*: Cy¹, AAB69318; Cy², AAB69319; P2, AAB69317; *Solanum tuberosum*: Cy, CAA52442; P1, CAA58775; P2, CAB52708; *Spinacia oleracea*: P1, CAA03939; *Triticum aestivum*: Cy, BAA97663.

(b) Partial protein alignment of Arabidopsis G6PDHs. Numbers preceding each row indicate: 1, G6PD1; 2, G6PD2; 3, G6PD3; 4, G6PD4; 5, G6PD5; 6, G6PD6. A second set of numbers represents the amino acid positions. Identical amino acids are indicated by black boxes, similar ones are shaded gray. Asterisks indicate cysteine residues conserved among plastidic predicted isoforms. The bar indicates the G6PDH active site motif DHYLGKE as defined by the PROSITE database.

genes annotated as encoding G6PDHs are available in GenBank and were previously designated *AtG6PD1–AtG6PD6* (for accession numbers refer to Figure 1 legend). A search for additional G6PDH sequences using portions of the amino acid sequences of *bona fide* G6PDHs did not yield new potentially G6PDH encoding genes for the *Arabidopsis* or rice genomes. The classifications of translated proteins in the phylogenetic tree (Figure 1a) follow those designated by Wendt *et al.* (1999), plastidic (P1, P2, and other), and cytosolic (Cy). The relationship among the three already described isoforms from *Arabidopsis* (G6PD1, G6PD3, and G6PD5) was in agreement with a previous report (Wendt *et al.*, 1999). Isoforms G6PD2 and G6PD6 were found to group with P2 and cytosolic isoforms respectively, each with highest similarity to another *Arabidopsis* isoform. Interestingly, G6PD4 did not group with any of the other *Arabidopsis* isoforms, but with a rice isoform.

We used ChloroP (Emanuelsson *et al.*, 1999) and TargetP (Emanuelsson *et al.*, 2000) which allowed us to evaluate the presence of a transit peptide in the translated protein and to predict the subcellular localization of all plant G6PDH isoforms. The predictions correlated with the phylogenetic clades not only for *Arabidopsis* but for all species examined. Translated proteins for *AtG6PD1*, *AtG6PD2*, *AtG6PD3*, and *AtG6PD4* were predicted as plastidic while those for *AtG6PD5* and *AtG6PD6* were cytosolic (Table 1), consistent with previous results (Kruger and von Schaewen, 2003).

The amino acid sequences of *Arabidopsis* isoforms showed high conservation with similarities ranging from 41 to 92% (data not shown). The cysteine residues reported to be conserved among plant plastidic isoforms that confer redox sensitivity (Wenderoth *et al.*, 1997; Wendt *et al.*, 1999) were also present in all isoforms predicted to be plastidic (Figure 1b). The G6PDH active site consensus sequence DHYLGKE as given by the PROSITE database (Hofmann *et al.*, 1999) was conserved in all except G6PD4, which differed in three of seven amino acids (Figure 1b). Interestingly, the most similar rice isoform also had two amino acids changed in the active site in the same positions as in G6PD4 of *Arabidopsis* (data not shown).

Table 1 The *Arabidopsis* G6PDH gene complement

Locus	Localization	a.a.	TP	MW (kDa)	Designation
At5g35790	plastidic	626	50	60	G6PD1
At5g13110	plastidic	646	50	62	G6PD2
At1g24280	plastidic	656	57	61	G6PD3
At1g09420	plastidic	674	49	65	G6PD4
At3g27300	cytosolic	516		59	G6PD5
At5g40760	cytosolic	515		59	G6PD6

Amino acids (a.a.) correspond to the pre-processed protein. Subcellular localization, transit peptide (TP) length in amino acids, and molecular weight (MW) of the mature protein were predicted *in silico*.

Biochemical characterization of recombinant proteins

The cDNAs for the mature G6PDH isoforms were expressed adding additional amino acids to accommodate a C-terminal *Strep*-tag that allowed subsequent purification from *Escherichia coli*. Previously, the same system was successfully used for the expression of potato G6PDHs (Wendt *et al.*, 2000). Recombinant protein was purified by affinity chromatography after which one major band was observed by SDS-PAGE (data not shown). The yield of individual recombinant proteins varied among isoforms and individual preparations from 0.5 to 3 mg l⁻¹ (protein/culture). All preparations lost activity rapidly upon storage at 4°C, -20°C with or without 20% glycerol, or -80°C following snap-freezing (data not shown). Therefore all experiments described here were carried out immediately following purification. All experiments were repeated two to three times. In separate experiments of an isoform, K_{mNADP^+} (K_m) values were reproducible whereas V_{max} values were lower when the time taken for purification process was long, or the column had been used repeatedly. In Figure 2(a–e) an optimal repeat experiment for each isoform is shown in which the highest V_{max} was obtained. All isoforms had a similar pH optimum of 8–8.5 (data not shown), and utilized NADP⁺ but not NAD⁺ in a concentration-dependent manner. In all cases the plot of specific activity against substrate concentration followed a hyperbolic function. Isoform G6PDH4 exhibited very low, but measurable NADP⁺ concentration-dependent activity indicating that NADP⁺ is indeed a substrate (data not shown). The low activity was likely due to the G6PDH-atypical amino acid difference in the conserved active site (Figure 1b) and thus, no further biochemical characterization of this isoform was pursued. Values for V_{max} were specific for each isoform but varied greatly among the different recombinant proteins ranging from 0.1 to approximately 100 U mg⁻¹ protein. All isoforms tested showed a concentration-dependent competitive inhibition by NADPH (Figure 2a–e), as determined by Lineweaver-Burk plots (data not shown). A mixed inhibition mode might be suggested from the changing values of apparent V_{max} for G6PD5 (Figure 2d). However, this is possibly due to experimental limitation in completely saturating the enzyme with the substrate. Moreover, Lineweaver-Burk plots of repeated experiments indicated competitive inhibition (data not shown). The biochemical characteristics of the recombinant enzymes are summarized in Table 2. The different isoforms had K_m values for NADP⁺ within the range of 1–20 µM, except for isoform G6PD6, which had a considerably higher K_m of 6.5 mM (Figure 2e, Table 2). The application of high substrate concentrations to this isoform, and addition of NADPH resulted in a high-assay background, and therefore K_{iNADPH} (K_i) was not determined for G6PD6. For any given isoform, a small K_i compared with the K_m is indicative of a tight feedback inhibition by NADPH. Only the K_i of G6PD2 was significantly

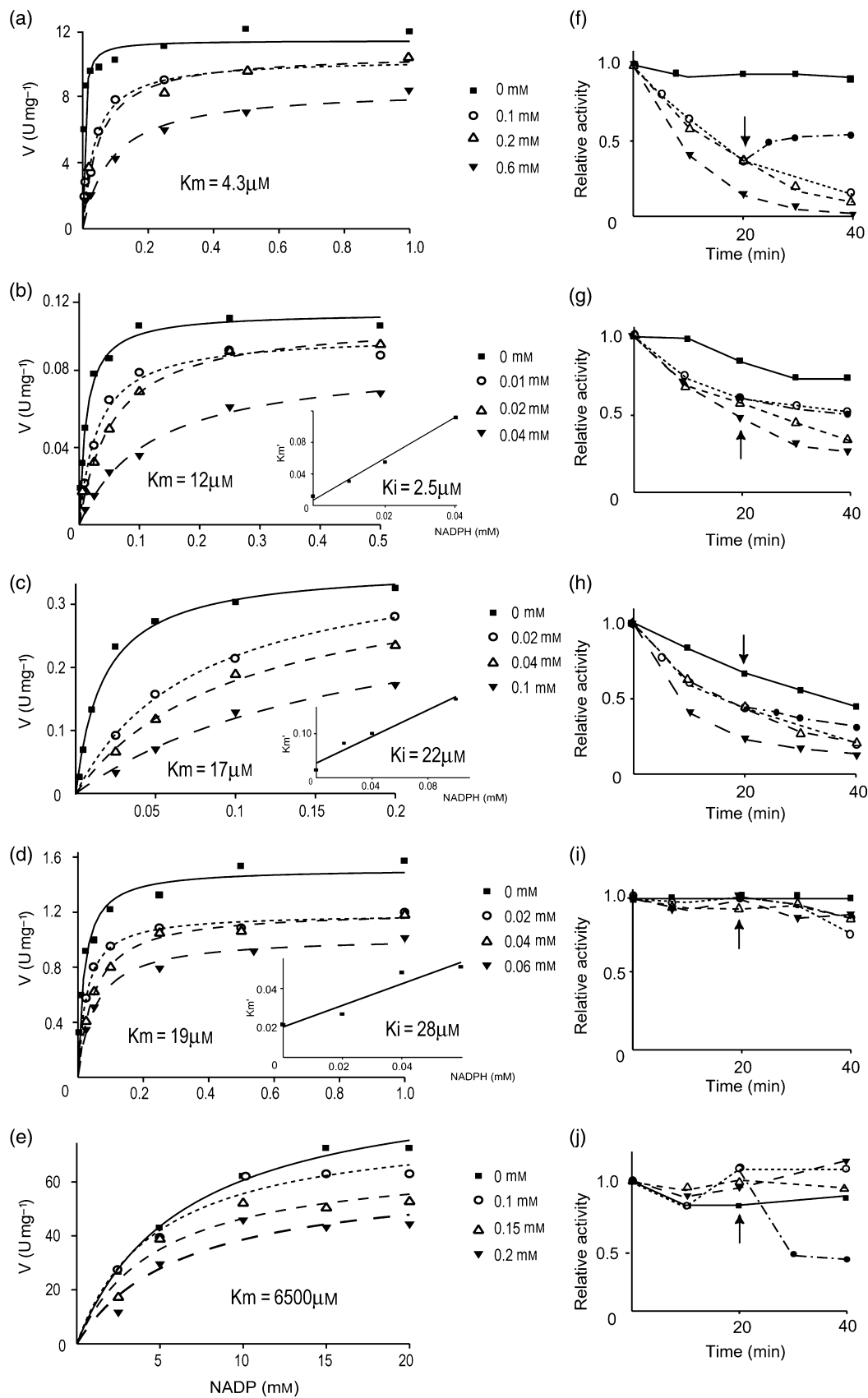


Table 2 Biochemical characteristics of recombinant G6PDH isoforms

Isoform	pH optimum	$V_{\max}$ (U mg ⁻¹)	K_m NADP ⁺ (μM)	K_i NADPH (μM)
G6PD1	8	11 (0.3)	4.3 (0.1)	30–70
G6PD2	8.5	0.11 (0.01)	12 (1)	2.5 (2)
G6PD3	8–8.5	0.36 (0.01)	17 (2)	22 (11)
G6PD4	8.5	<0.04	ND	ND
G6PD5	8.5	1.5 (0.05)	19 (3)	28 (14)
G6PD6	8	100 (5)	6500 (900)	ND

Values shown in parenthesis indicate standard deviation. Experiments were repeated two or three times and values were derived from experiments shown in Figure 2. ND, not defined.

lower than its K_m , whereas the two kinetic constants were comparable for G6PD3 and G6PD5 (Figure 2, Table 2). A precise K_i value could not be obtained for G6PD1, but results from several experiments were reproducibly in the range of 30–70 μM (Table 2). This was an order of magnitude higher than its K_m (4.5 μM) suggesting that G6PD1 is less prone to feedback inhibition. Taken together, these results indicate the different sensitivities to NADPH for each isoform, which may reflect the differential regulation required in different photosynthetic and redox environments, where the respective isoforms are present.

Plastidic isoforms are inactivated by reduction whereas cytosolic isoforms show varying sensitivities

For the five active isoforms, the sensitivity to dithiothreitol (DTT) which reduces disulfide bonds to dithiol (Johnson, 1972; Lendzian, 1980) was examined (Figure 2f–j). The plastidic isoforms exhibited concentration-dependent inactivation by DTT over time (Figure 2f–h). This inactivation was reversed by addition of sodium tetrathionate (NaTT) only for G6PD1 (P1 isoform), although the activity did not completely recover (Figure 2f). For the P2 isoforms G6PD2 and G6PD3 a significant reactivation by NaTT was not observed. This could be due to an irreversible inactivation by DTT as was reported for the potato P2 isoform, which was reactivated only when incubated with DTT and thioredoxin but not with DTT alone (Wendt *et al.*, 2000), or it could be related to decreased stability during longer incubation times required for this experiment (Figure 2g,h). The inactivation by reduction is in agreement with previous reports for G6PDH isoforms from potatoes, which showed that only the plastidic isoforms possess the conserved cysteine residues and are

sensitive (von Schaewen *et al.*, 1995; Wenderoth *et al.*, 1997; Wendt *et al.*, 2000). The cytosolic isoforms G6PD5 and G6PD6 were not affected by DTT (Figure 2i,j). Application of the oxidant NaTT to G6PD6 resulted in inactivation (Figure 2j), which was irreversible (data not shown), whereas G6PD5 was not affected by either DTT or NaTT (Figure 2i).

Expression of G6PDH genes varies greatly in different tissues

The frequency of reported expressed sequence tags (ESTs) in different cDNA libraries was examined for each isoform (as of July 2004) (Figure 3a). Isoform *AtG6PD6* was the most represented with 24 ESTs and was the only isoform reported to be present in many different library sources including whole plant, seedling, silique, seed, flower, and root. All other isoforms had ESTs in libraries from whole plants, seeds, or siliques. This *in silico* analysis allowed us to estimate the abundance of transcripts derived from each gene but the tissue-specificity for the isoforms was still unknown except for *AtG6PD6*. Therefore we examined the mRNA abundance for each isoform in different tissues using RT-PCR.

The primers were designed to amplify specific 3'-UTRs for each isoform. The primer pairs for each isoform were optimized and tested for similar efficiencies with genomic template (Figure 3b), allowing comparisons of transcript abundance between different isoforms. In Figure 3(c), the gel photos are intended to allow comparisons of mRNA abundance across different isoforms whereas the graphs show relative levels of mRNA of a single isoform normalized to actin in different tissues. Judging from the intensity of amplified PCR products detected on the gel, overall the mRNA was most abundant for *AtG6PD6* (Figure 3c). In specific tissues mRNA was abundant for *AtG6PD1*, *AtG6PD2*, *AtG6PD3*, and *AtG6PD5* (Figure 3c). The mRNA for *AtG6PD4* was detectable but its abundance was considerably lower compared with other isoforms (Figure 3c). This finding together with the amino acid difference in the active site and lack of activity suggests that *AtG6PD4* does not encode for a functional G6PDH. The mRNA of the highly expressed *AtG6PD6* gene encoding a predicted cytosolic isoform was found with similar levels in all tissues examined, suggesting it to encode the major G6PDH isoform throughout the plant, whereas the mRNA from *AtG6PD5* predicted to encode the other cytosolic isoform was detected at high level in leaves. The mRNA of the plastidic P1 isoform encoded by *AtG6PD1*

Figure 2. Kinetic parameters and the effect of reduction.

(a–e) Substrate (NADP⁺) saturation curves under various inhibitor (NADPH) concentrations. The NADPH concentrations used for each enzyme are shown with their symbols accompanying their saturation curves. Inserts show the plots of apparent K_m (K_m') under different concentrations of the inhibitor. (f–j) Time-dependent change in G6PDH activity following incubation with various concentrations of DTT. Activity is described as relative value starting with 1 at time zero. Arrows indicate the time when NaTT was added. Symbols indicate different concentrations of DTT and NaTT added, solid square, 0 mM DTT; open circle, 3 mM DTT; open triangle, 5 mM DTT; reverse solid triangle, 10 mM DTT; solid circle, 5 mM NaTT added to 3 mM DTT. (a, f) G6PD1; (b, g) G6PD2; (c, h) G6PD3; (d, i) G6PD5; (e, j) G6PD6.

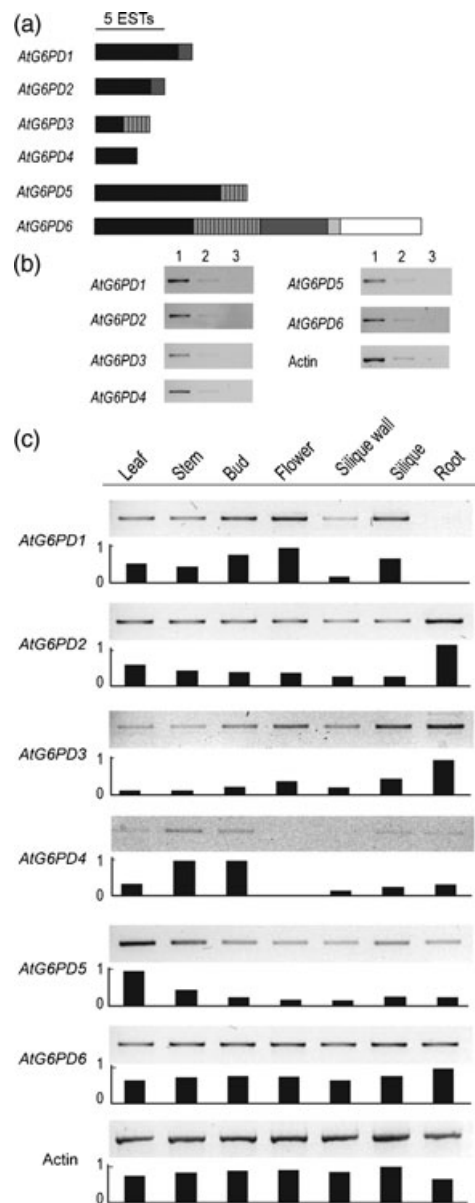


Figure 3. Expression of G6PDH isoforms in various tissues. (a) Number of ESTs reported. Bars indicate different sources of EST libraries. Solid, whole plant; striped, green siliques; dark gray, developing seeds; light gray, flowers; open, roots. (b) Efficiency of PCR primers tested in serial dilutions of genomic DNA template. PCR performed with genomic DNA template (lane 1); 1/10 dilution (lane 2); 1/100 dilution (lane 3). (c) RT-PCR from various tissues. Bar graphs indicate relative intensities in different tissues normalized to the actin fragment intensity from the respective tissue. The highest normalized intensity obtained in a certain tissue was defined as 1.

was present in all tissues except for roots. Higher levels of mRNA were found in developing tissues such as buds, flowers, and early siliques compared with photosynthetic tissues (leaves and stems). This expression pattern was similar to that for the potato P1 isoform, for which the mRNA

was predominant in green tissues (Wendt *et al.*, 2000). On the contrary, mRNAs for the P2 isoforms encoded by *AtG6PD2* and *AtG6PD3* were detectable in all tissues but their abundance was considerably higher in roots. *AtG6PD2* and *AtG6PD3* are induced in roots and shoots by nitrate (Wang *et al.*, 2003). In potato, the P2 gene was also induced under N-sufficiency in the roots when grown in hydroponic cultures, although expression was ubiquitous in soil-grown potato plants (Wendt *et al.*, 2000).

Three major isoforms of G6PDH activity are detected *in vivo*

As shown in Figure 2, G6PDH enzymatic activity was regulated post-translationally by different factors *in vitro* (substrate and inhibitor concentrations, and reductive inactivation). To explore the roles for each isoform *in vivo*, we examined the activity of individual isoforms in protein extracts by separating native proteins based on their isoelectric point and by detecting their activities rather than protein amounts, because enzyme activity can be considered the final level of gene expression. Zymograms are classic tools in the studies of G6PDH polymorphisms and isozyme identifications (Eanes, 1983; Steele *et al.*, 1968). On a given gel, higher mobility of a band indicates a lower isoelectric point of the protein. A bacterial G6PDH was loaded as a positive control. Three bands representing G6PDH activity (isoforms) were detected in stems and roots, of which two were also present in leaves (Figure 4a, solid arrowheads, A, B, C). Three common isoforms (two more and one less active) with lower mobility were detected in buds, flowers, siliques, and seeds (Figure 4a, open arrowheads, A', B', C'). No activity was detected when G6P was excluded (data not shown). With similar amounts of total protein loaded, of all tissues roots had three highly active isoforms (Figure 4a, A, B, C), whereas in photosynthetic leaves and stems, activity staining was less intense (Figure 4a, A, B) suggesting a significant difference in total G6PDH activity per total protein in these tissues. Developing tissues such as buds, siliques, and seeds had the two isoforms with higher mobility showing comparatively higher activity (Figure 4a, B', C'), whereas flowers had most activity in the second isoform (Figure 4a, B'). In all developing tissues, the lowest mobility isoform (Figure 4a, A') was low in activity. In a separate experiment, less concentrated tissue extracts (approximately 1.7 mg protein ml⁻¹) were assayed for G6PDH activity by zymograms and by measuring the total G6PDH activity in cell extracts spectrophotometrically (Figure 4b). The specific activity showed good correlation with band intensities observed on zymograms, indicating that this method is quantitative (Figure 4b, compare zymogram and graph). In addition, the proteins detected in this control experiment had similar mobility across different tissues, suggesting that the shift in bands observed in Figure 4(a) (A, B, C and A', B', C') is due to high sample

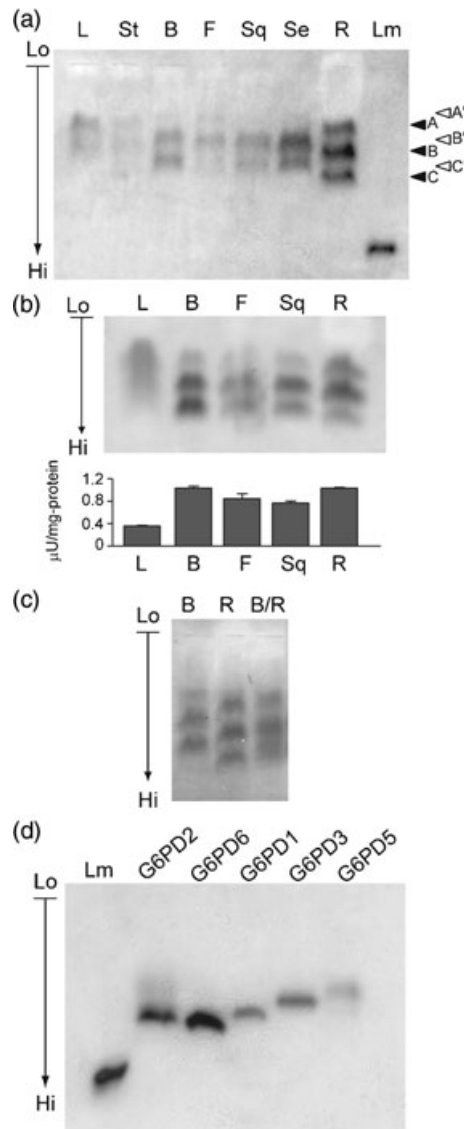


Figure 4. G6PDH isoform activity detected by zymograms. G6PDH activity was detected by zymogram from (a–c), wild-type tissue; (d), purified recombinant G6PDHs. (a), concentrated extracts; (b), dilute extracts also assayed spectrophotometrically as shown in the graph; (c), dilute and mixed extracts. Lanes: L, leaves; St, stems; B, buds; F, flowers; Sq, siliques; Se, seeds; R, roots; Lm, G6PDH from *Leuconostoc mesenteroides* (positive control); B/R, mixed buds and roots. Solid and open arrowheads indicate positions of bands detected in L, St, R (A, B, C) and B, F, Sq, Se (A', B', C'), respectively. Direction of electrophoresis is indicated by the arrow, the start of which is the origin. Lo, Hi, low and high protein mobility, respectively.

concentrations (approximately $6.3 \text{ mg protein ml}^{-1}$) applied to obtain the clearest possible banding pattern on the zymogram. Furthermore, to independently address whether the two sets of three bands with shifted mobility visible in Figure 4(a) were representative of the same three isoforms, extracts from wild-type buds and roots were mixed and analyzed (Figure 4c). The resulting zymogram showed an intermediate banding pattern. This, together with the

zymogram pattern of less concentrated extracts (Figure 4b), strongly suggests that isoforms detected as A, B, C are presumably equal to A', B', C' in Figure 4(a). Because some isoforms were observed to be more labile as recombinant enzymes (G6PD2 and G6PD3, Figure 2g,h), it was considered that they were preferentially lost under zymogram conditions. To address this possibility, the stability of the individual recombinant enzymes was analyzed by zymograms (Figure 4d). Similar amounts, approximately 0.2–1 mU, for each isoform were applied. All isoforms were detected except for G6PD4 (data not shown) indicating that none was significantly less stable under these conditions. Different extraction methods were tested, for example, excluding DTT or protease inhibitors from the extraction buffer. These did not result in changes visible on the zymograms (data not shown) indicating that the lack of detection of other isoforms is not due to inactivation by DTT or protease inhibitors during extraction.

Identification of the genes encoding active G6PDH isoforms in vivo

To identify the genes coding for the active isoforms detected on zymograms, we took advantage of T-DNA insertion lines obtained from the SALK institute (Alonso *et al.*, 2003). The informative SALK lines available and their confirmed T-DNA insertion sites are shown in Figure 5(a). The homozygous T-DNA insertion lines for the respective genes will be referred to as *g6pd5* and *g6pd6*. The 3'-UTR transcripts of *AtG6PD5* and *AtG6PD6* were lost in leaves of *g6pd5* and *g6pd6* by RT-PCR (data not shown). To cover the full range of banding patterns observed (Figure 4a), protein was extracted from leaves (Figure 4a, A, B), buds (Figure 4a, A', B', C'), and roots (Figure 4a, A, B, C) from wild-type, *g6pd5*, and *g6pd6*, and was analyzed by zymograms. Protein extracts from *g6pd6* showed two isoforms, lacking the highest mobility isoform band present in wild-type buds and roots, leading to a parallel increase in intensity of a low mobility isoform in all tissues examined (Figure 5b, compare WT and *g6pd6*). This result indicates that (i) the isoform detected as the highest mobility band in wild-type plants that was lost from its original relative position in *g6pd6* is a product of *AtG6PD6*, and (ii) although detected with different mobilities in Figure 4(a), band C' in buds and band C in roots represent the same protein encoded by *AtG6PD6*. The isoform with higher mobility than that represented by an intensified band seemed reduced in activity especially in root tissues (Figure 5b, *g6pd6* lane R), which could be partly due to the difficulty in its separation from the highly active isoform. Similarly, *g6pd5* lacked the isoform with the lowest mobility in all three tissues compared with wild type, although residual activity was detected in leaves suggesting this T-DNA insertion line is an incomplete knockout (Figure 5b, compare WT and *g6pd5*, lane L). This result indicates that

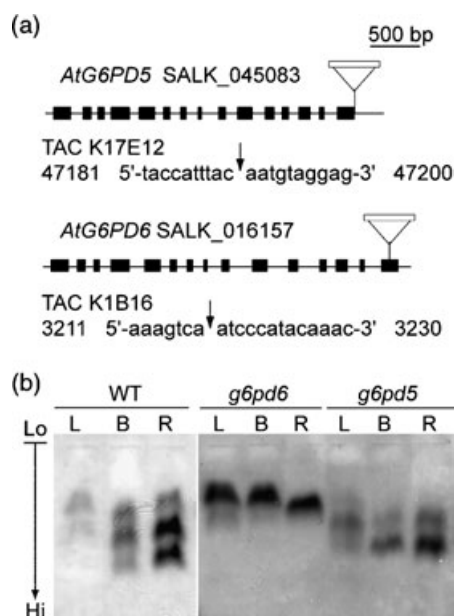


Figure 5. Identity of G6PD5 and G6PD6 on zymograms. (a) Schematic gene structure of *AtG6PD5* and *AtG6PD6*. The T-DNA insertion sites are shown as the nucleotide position in the respective genomic BAC clones [GenBank accession numbers, AP000381 (K17E12); AB015470 (K1B16)]. Black boxes indicate the coding region. (b) G6PDH activity detected by zymogram in tissues from wild type (WT) and T-DNA insertion plants (*g6pd5* and *g6pd6*). Lanes: L, leaves; B, buds; R, roots.

the lowest mobility bands that were lost from all tissues of *g6pd5* were due to proteins encoded by *AtG6PD5*. The loss of the low-mobility isoform was accompanied by an increase in intensity of a high-mobility isoform with a similar mobility as G6PD6 in all three tissues examined (Figure 5b, under *g6pd5*). The question arose whether G6PD6 and G6PD5 activities are reciprocally induced in *g6pd5* and *g6pd6*, respectively. The mobilities of the induced isoforms were similar (if not overlapping) to G6PD5 (lowest mobility) and G6PD6 (highest mobility), which suggests that in *g6pd5* and *g6pd6*, the isoforms G6PD6 and G6PD5 were induced, respectively. Nonetheless, it cannot be ruled out that an isoform that was previously not detected with the same mobility was induced in *g6pd5* and *g6pd6* or that the T-DNA insertion generated an active protein with altered structure that migrated together with G6PD5 and G6PD6 in both *g6pd6* and *g6pd5*, respectively. However these latter possibilities seem rather unlikely, as one would statistically expect a different migration for an altered or different protein in at least one of the two T-DNA insertion lines.

The results from multiple experiments suggest that the activity of only three isoforms were detected on the zymogram, which leaves one isoform, presumably a plastidic isoform, to be identified. Plants homozygous for a T-DNA insertion in an exon of *AtG6PD3* did not lose any activity visible on the zymogram although the 3'-UTR transcript of

the gene was aberrant (data not shown). Preferential instability of this isoform on zymograms can be ruled out as well, as shown in Figure 4(d). These results suggest that this isoform has little or no activity *in vivo* even in roots where it is predominantly expressed under normal conditions (Figure 3c). However, these results do not rule out its conditional role as the gene coding for the G6PD3 is induced by nitrate in Arabidopsis (Wang *et al.*, 2003). Plants homozygous for T-DNA insertions in regions several hundred base pairs upstream of *AtG6PD1* and *AtG6PD2* were isolated and analyzed but did not result in the loss of transcripts (data not shown). Therefore the gene corresponding to the isoform that migrates in between G6PD5 and G6PD6 remains uncertain.

Discussion

The G6PDH complement in rice and Arabidopsis

The phylogenetic analysis gave new information on G6PD2 and G6PD6 belonging to P2 and cytosolic groups, respectively. We also showed that the rice genome has loci presumably encoding for five G6PDHs, which represented members from each group as in Arabidopsis. Isoform G6PD4 has amino acid changes in the otherwise conserved active site, little activity *in vitro*, and low gene expression *in vivo*, properties suggesting that it is a non-functional G6PDH. There is a potential ortholog in rice that also has amino acid changes in the active site. Two of these positions are the same within the two proteins, although the replaced amino acids are different (data not shown). Both have plastidic G6PDH characteristics, N-terminal transit peptides and conserved cysteine residues, making it possible that these isoforms were once active plastidic G6PDHs. Whether G6PD4 requires other cofactors or has activity different from typical G6PDH enzymes is not known at this time.

Cytosolic G6PDH isoforms of Arabidopsis

The characterization of G6PDH isoforms is summarized in Table 3. The two cytosolic isoforms have distinct characteristics despite their similarity (92%, data not shown). The abundance of mRNA and *in vivo* activity suggest that G6PD6 is the major cytosolic isoform of G6PDH in Arabidopsis. Isoform G6PD6 is unique with both V_{max} (100 u mg^{-1}) and K_m (6.5 mM) values being at least an order of magnitude greater than those of the other isoforms. The cytosolic NADP⁺ concentration in pea leaves in dark or light has been reported to be 140 μM , and a similar concentration of NADPH has been observed (Igamberdiev and Gardestrom, 2003). This suggests that under physiological conditions, G6PD6 is greatly sub-saturated and responds linearly to the change in NADP⁺, that is, redox levels, whereas the specific activity of the other cytosolic isoform, G6PD5, will not change (Figure 6a).

Table 3 Summary of the *Arabidopsis* G6PDH isoform properties

Group	Gene	Expression	Feedback inhibition	Reduction	<i>In vivo</i> activity ^a
Cy	<i>AtG6PD5</i>	Prevalent in leaf	$K_m \sim K_i$	Insensitive	Leaf and root
	<i>AtG6PD6</i>	Ubiquitous	ND	Insensitive	Ubiquitous except leaf
P1	<i>AtG6PD1</i>	Higher in developing organs Absent in root	$K_m < K_i$	Sensitive	ND
P2	<i>AtG6PD2</i>	Highest in root	$K_m > K_i$	Sensitive	ND
	<i>AtG6PD3</i>	Highest in root	$K_m \sim K_i$	Sensitive	Not detected
Other	<i>AtG6PD4</i>	Low	ND	ND	ND

^aPresence was determined by loss of a band in KO compared with wild type.

Cy, cytosolic; P1, plastidic P1; P2, plastidic P2; $K_m > K_i$, $K_m \sim K_i$, feedback inhibited; $K_m < K_i$, no feedback inhibition; ND, not defined.

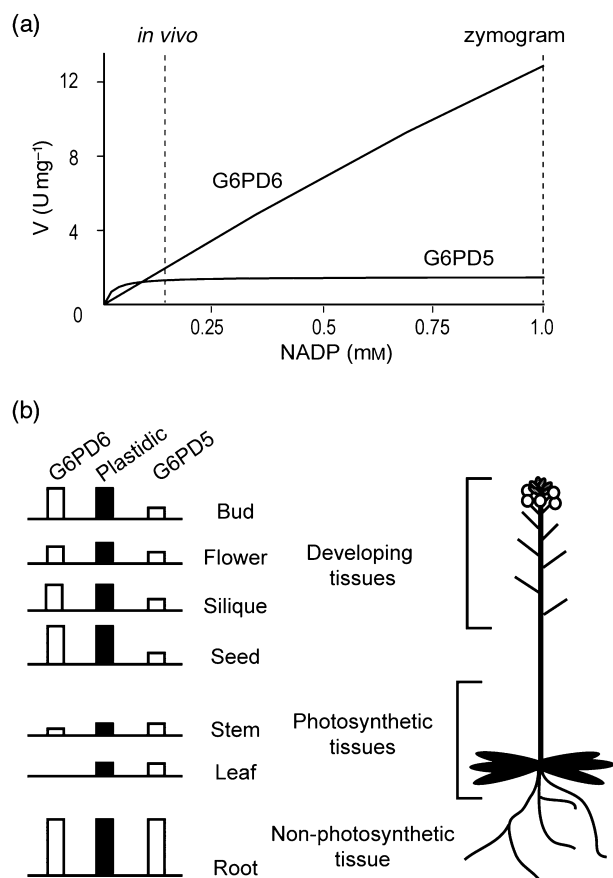
Isoform G6PD5 is the major cytosolic isoform in leaves of wild-type plants, and its activity is very low in other tissues. The specific role of G6PD5 which appears to replace G6PD6 only in leaves is not known. The fact that G6PD5 is tolerant to

both reduction and oxidation while G6PD6 is inactivated by oxidation (Figure 2i,j) may allow this isoform to carry out unique functions in leaves.

The activities of the two cytosolic isoforms appear to be coordinated. In wild-type plants, all tissues except roots had only one of the two cytosolic activities (Figure 4a), with G6PD5 being detected in leaves and G6PD6 in other parts of the plant. A reciprocal, but not equal, induction of their activities was observed in the two T-DNA insertion lines, assuming that the two induced isoforms we observed in *g6pd5* and *g6pd6* were G6PD6 and G6PD5, respectively (Figure 5b). No induction at the mRNA level was detected in the T-DNA insertion lines (data not shown). It is not known whether this coordination and regulation occurs at the level of enzyme amount, enzyme activity, or post-translational modification. Cooperation between cytosolic and plastidic OPPP has been observed in maize mutants lacking cytosolic OPPP (Averill *et al.*, 1998). Increased rates of flux through OPPP were observed in these lines in response to nitrate, although both genes encoding cytosolic 6-phosphogluconate dehydrogenase were disrupted. We observed an increase in activity of what appeared to be the remaining cytosolic isoform when one was disrupted, but none of the activities related to the plastidic G6PDH isoforms was stimulated. Whether an increase in plastidic G6PDH induction occurs when both cytosolic G6PDH isoforms are disrupted remains to be seen. Nonetheless our results suggest that cytosolic G6PDHs are coordinated in their activity, at least to an extent, presumably to maintain the redox balance in the cell.

Plastidic G6PDH isoforms in *Arabidopsis*

The P2 isoform encoding genes *AtG6PD2* and *AtG6PD3* showed similar abundance of ESTs, expression levels, and patterns; lower in photosynthetic tissues and highest in roots (summarized in Table 3). Their proteins had similar kinetic parameters V_{max} , K_m , although G6PD3 may be less sensitive to feedback inhibition. In contrast, *AtG6PD1*,

**Figure 6.** G6PDH activities in *Arabidopsis*.

(a) Schematic representation of activities of the two cytosolic isoforms under a range of predicted concentrations of cytosolic $NADP^+$. The dotted line indicates the $NADP^+$ concentration reported for the cytosol (0.14 mM) and that used for the zymogram (1.0 mM).

(b) Schematic representation of relative activity of isoforms in plant tissues detected in zymograms.

belonging to the P1 group is highly expressed in developing tissues, to a lesser extent in green tissues, and absent in roots (Figure 3c, Table 3). The kinetic parameters for the G6PD1 protein were unique, especially its reduced tendency to feedback inhibition (Table 2). All plastidic isoforms were sensitive to reductive inactivation by DTT, although the degree of sensitivity could not be directly compared due to their lability at room temperature (Figure 2f–h). The activity of the plastidic isoforms was measured with no prior treatment. We cannot entirely rule out the possibility that the recombinant plastidic enzymes as prepared were not completely oxidized and fully active. However, because incubation with NaTT caused only a marginal activation (data not shown), we conclude that a large portion (if not all) of the protein was purified in its most active state.

Previous studies in potato and barley supported the view that P1 isoforms, which are predominantly present in green tissues, are feedback inhibited by NADPH, whereas ubiquitous P2 isoforms are less sensitive to feedback inhibition and to reductive inactivation (Esposito *et al.*, 2003; Wendt *et al.*, 2000). Our results show an alternative case in Arabidopsis where G6PD1 (P1) is the least NADPH-sensitive of all isoforms and its gene was ubiquitously expressed except for roots, while *AtG6PD2* and *AtG6PD3* (P2) expression was not ubiquitous, but prevalent in roots (Figure 3c, Table 3). There are some possible explanations for these species-specific differences. The model proposed by Wendt *et al.* (2000) assumes the presence of two ancestral G6PDHs that arose to have different roles and characteristics in different tissues (P1 and P2), and predicts that similar features are maintained among the P1 and P2 isoforms in different plants. As expression patterns are more often determined by regulatory sequences outside of the coding region, which is not reflected in the phylogenetic tree, it is possible that in the case of Arabidopsis G6PDHs, changes in regulatory sequences occurred that altered expression patterns. Subsequently this was accompanied by amino acid changes leading to altered tertiary structures that rendered the enzymes more suitable for their roles in new tissues. In addition, we have shown that there is little correlation between mRNA levels and activity of G6PDH isoforms, which may indicate that expression patterns are not strong indicators of the *in vivo* roles of the native gene products at least in the case of G6PDH isoforms.

Plastidic isoforms *AtG6PD1* and *AtG6PD2* are the likely candidates for the presumed ubiquitous plastidic isoform detected in zymograms (because T-DNA insertion in *AtG6PD3* did not lead to activity loss on a zymogram and G6PD4 is inactive). The overall transcript abundance throughout the plant was similar for both, but in neither case the expression patterns matched the activity present in all tissues as detected by zymograms. These results also suggest that the major regulatory mechanisms of Arabidopsis G6PDH isoform activity are post-transcriptional, at the

levels of protein amount, enzyme activity and post-translational redox modification. In potato, the induction of cytosolic G6PDH activity and increased mRNA abundance was observed in response to sugar while in contrast, plastidic (P1) activity was induced by oxidative conditions, although protein levels remained the same (Hauschild and von Schaewen, 2003).

We initially detected in zymograms two sets of three apparent isoforms whose migration was slightly shifted (Figure 4a), two of which were identified to be pairs of G6PD5 (Figure 4a, A and A') and G6PD6 (Figure 4a, C and C') using T-DNA insertion plant extracts. Our results suggest that bands B and B' in Figure 4(a) represent one G6PDH isoform, hence there remains only one isoform to be identified on zymograms. We favor the possibility of G6PD1 being the unidentified plastidic isoform visible on zymograms for the following reasons: (i) its $V_{\max}$ was relatively high as for the two cytosolic isoforms for which activity was detected on zymograms, and (ii) its reduced tendency to feedback inhibition allows this enzyme to be active in photosynthetic tissues (Table 2). The activity of G6PD3, a P2 isoform, was not detected in the plant under normal conditions in agreement with its relatively low expression and $V_{\max}$ of the recombinant protein. Nonetheless, the possibility of G6PD3 and/or G6PD2 being active under specific conditions remains as *AtG6PD2* and *AtG6PD3* are induced by nitrate in Arabidopsis (Wang *et al.*, 2003). Furthermore, the correlation of activity of a P2 isoform with oxidative stress has been shown in a recent study, in which transgenic tobacco plants expressing anti-sense cDNA for the plastidic P2 isoform unexpectedly showed increased tolerance to methylviologen (Debnam *et al.*, 2004). Whether Arabidopsis P2 isoforms are involved in oxidative or other stress responses is not known at this time.

Interplay of different G6PDH isoforms in vivo

It has been widely accepted that native forms of G6PDH occur in homodimers or homotetramers, the latter shown to be the active form in maize leaves (Kahler and Kirkman, 1983; Srivastava and Anderson, 1983; Valenti *et al.*, 1984). Thus we assumed that the native active G6PDH enzymes are homomers, presumably in the form of homotetramers. This was supported by the activity of the recombinant proteins expressed from a single gene in *E. coli*. We also observed the reduction of only one band of defined mobility (i.e., isoelectric point) by gene disruption indicating that G6PD5 and G6PD6 are involved in a single complex for which activity could be detected on a zymogram.

The NADP⁺ concentration (1 mM) applied to the zymograms is higher than *in vivo* concentrations, 0.14 mM in the cytosol (Igamberdiev and Gardestrom, 2003), and 0.4 mM total NADP(H) in the chloroplast stroma (Igamberdiev *et al.*, 2001). Therefore maximal activity is detected for all isoforms

except for G6PD6, for which the K_m is higher (6.5 mM), a fact which leads to the over-representation of this isoform on a zymogram compared with other isoforms (Figure 6a). Nonetheless, our zymogram experiments revealed many new findings such as the different levels of total G6PDH proteins present in active forms in a tissue as well as distinct combinations of isoforms. In summary, we observed three patterns of active isoforms as follows (Figure 4a and depicted in Figure 6b): (i) isoforms G6PD5 and a plastidic isoform in green tissues, (ii) G6PD6 and a plastidic isoform in developing tissues (mostly a plastidic isoform in flowers), and (iii) all three isoforms in non-photosynthetic roots. Although G6PDH has conventionally been described as a housekeeping enzyme or dark enzyme, these results suggest that no single isoform is capable of all functions and that each has its individual roles. These patterns and combinations perhaps reflect the metabolism of the subcellular compartments and photosynthetic activity of the tissue, suggesting that G6PDH activity patterns may infer metabolic states brought about by differences in rates of reductive and oxidative reactions and the demand for NADPH.

Conclusions and perspectives

In this post-genomic era the deciphering of genomic sequences is routinely accomplished and gene expression at the mRNA level can be readily determined using multiparallel approaches. However, the parallel activity analysis of multiple enzymes still presents great challenges, even if the enzymes in question catalyze similar or identical reactions. This case study showed that none of the isoforms of G6PDHs in *Arabidopsis* were exactly the same in activity, sensitivity to inhibitors or with regard to the expression of the respective genes. As G6PDHs are enzymes of central metabolism, what can be expected for the much larger enzyme families of secondary metabolism, which might have highly specialized functions? The difficulty in the comparison of gene family members and their gene products arises from the need to detect levels of mRNA, proteins, or enzyme activity with high specificity and equal sensitivity for each member to allow meaningful conclusions. For example, RT-PCR conditions need to be carefully calibrated for each set of specific primers, different anti sera against individual isoforms need to be highly specific and equally sensitive in Western blots, and enzyme isoforms need to be equally stable under the chosen assay conditions. With the exception of the generation of specific antisera, all the conditions were met in the current G6PDH study. Unlike the generation of specific antisera, a time-consuming process, zymograms are well suited for a genomic approach, because they provide parallel information and give an instant overview of the isoform activity for a given tissue or under a given environmental stimulus. However, they have to be carefully controlled and will not be applicable to all classes of

enzymes. Other potential pitfalls of a global enzyme analysis approach as described here arise from the fact that recombinant proteins with affinity tags are analyzed, which might differ in activity and sensitivity to regulators when compared with their native forms. Furthermore, many enzymes have a heteromeric subunit composition, are integral to membranes, or are part of larger complexes and their activity cannot be reconstituted from a single protein. While there is much to be learned from the systematic activity analysis of protein families, the technical feasibility and the appropriateness of biochemical approaches needs to be carefully evaluated in each case. Systematic activity analysis in parallel with global structural analysis of proteins will be required to more fully understand the functioning of plant cells. After all, gene expression becomes manifested in the biochemical activity of the encoded gene product, in many cases an enzyme involved in central or secondary metabolism.

Experimental procedures

Bioinformatics

All *Arabidopsis* and rice genes annotated as G6PDH as well as their coding sequences and all reported ESTs encoding *Arabidopsis* G6PDHs were searched through The Institute for Genomic Research website (<http://www.tigr.org/tdb/tgi/>). Subcellular localization of proteins was predicted using ChloroP (Emanuelsson *et al.*, 1999) and TargetP (Emanuelsson *et al.*, 2000). Transit peptides were removed from the proteins predicted to be plastidic and the similarities of the mature protein sequences were determined by ClustalW (Li, 2003). The phylogenetic tree was assembled with the aid of Biology Workbench (San Diego Supercomputer Center, University of California, San Diego; <http://workbench.sdsc.edu/>) and Phylip DrawTree (Felsenstein, 1989). Active consensus sites were identified based on the PROSITE database (Hofmann *et al.*, 1999) using MOTIF (<http://www.motif.genome.jp>).

Expression vector constructs and cloning

For tissues other than siliques total RNA was extracted with RNeasy (Qiagen, Valencia, CA, USA). Silique RNA was extracted as previously described (Yu *et al.*, 2004). Following DNase I treatment (Roche, Indianapolis, IN, USA), cDNA was synthesized from 300 ng of RNA using a reverse transcriptase kit (OmniScript; Qiagen). For cloning into expression vectors, regions coding for the mature G6PDH isoforms were amplified from RNA of siliques 5–8 days after flowering. From the RT reaction, 5% was used as template for subsequent PCR reactions (Hotstartaq; Qiagen) with 30 cycles. Primers as listed in Table 4 were designed to produce the mature proteins. The GenBank/EMBL accession numbers for the nucleotide sequences used for primer design of *Arabidopsis* G6PDHs are as follows: *AtG6PD1*, AY099561; *G6PD2*, AY065042; *AtG6PD3*, AY139768; *AtG6PD4*, AY056232; *AtG6PD5*, AJ010970; *AtG6PD6*, AJ010971. The PCR products for all genes except for *AtG6PD6* were sub-cloned into pCR2.1-TOPO (Invitrogen, Carlsbad, CA, USA), then cloned into the expression vector pASK-IBA3 (IBA, St Louis, MO, USA) using either *KpnI* or *EcoRI* sites. The orientation of the genes

Table 4 Primer sequences

Gene	Primer
<i>AtG6PD1</i>	p(+) 5'-GGGGTACCATTCTTCGCCGAGAAACATTC-3'
	p(-) 5'-CGGAATTCAGCTTCTCCAAGATCTCCCC-3'
	e(+) 5'-AAAGTTCCTGGTCTTGGAAAT-3'
	e(-) 5'-AATGCTCGATTAGAACCA-3'
<i>AtG6PD2</i>	p(+) 5'-CGGAATTCCTCCAGAGGAAGTCTGGTC-3'
	p(-) 5'-CCGGTACCCTGGTCAATACTGACGTAC-3'
	e(+) 5'-CCCTGGTTTAGGAATGAGAT-3'
	e(-) 5'-TAAGAGAAACCCCTTTGGTT-3'
<i>AtG6PD3</i>	p(+) 5'-GGGTACCTCGCAGAGGAGGTCCGTT-3'
	p(-) 5'-GGGTACCTGATCAAGACTTAGGTCTC-3'
	e(+) 5'-TCAAAGGAGATTCCAGATGC-3'
	e(-) 5'-CAAAAAGGGTGGCAAGAATA-3'
<i>AtG6PD4</i>	p(+) 5'-CGGAATTCCTCAACGAGGAGGCTCGA-3'
	p(-) 5'-CGGAATTCGTCATCTGCCCATGGGACA-3'
	e(+) 5'-CCGGAGTTGTATGAATTTGG-3'
	e(-) 5'-TTCGTCATCTTCTTCTCCA-3'
<i>AtG6PD5</i>	p(+) 5'-GGGGTACCCTGGGTTCTGGTCAATGGCA-3'
	p(-) 5'-GGGGTACCAATGTAGGAGGATCCAAAT-3'
	e(+) 5'-GAAGTGAATCGGTCCATA-3'
	e(-) 5'-CCACAAAGCTGTTTTGCATT-3'
<i>AtG6PD6</i>	p(+) 5'-GGGGTACCATGGGATCTGGTCAATGGCAGC-3'
	p(-) 5'-CCGGTACCTAGTGTAGGAGGGATCCAGATA-3'
	e(+) 5'-CAACCAGGTCTAGATATGAA-3'
	e(-) 5'-TGTCAAAGTGAACAGCTGAA-3'
<i>ACTIN-1</i>	e(+) 5'-AACAATCGATGGACCTGACTCG-3'
	e(-) 5'-TGCGACAATGGAAGTGAATGG-3'
<i>g6pd5</i>	t(+) 5'-GAAGTTTTTGGCTTTGCTGCG-3'
	t(-) 5'-GCAGTTATTCTGTGTGTATACGTCG-3'
<i>g6pd6</i>	t(+) 5'-CTAACTCTCTTGAACAGGTC-3'
	t(-) 5'-AAACACCAAACCTCGAGGAGGGA-3'
T-DNA LB ^a	t(-) 5'-GTTACGCTAGTGGGCCATCG-3'

^aModified from LBa1 primer suggested on SALK website (<http://signal.salk.edu>).

Primers used for cloning of recombinant protein production (p), gene expression analysis (e), and genotyping (t).

was confirmed by sequencing. The expression of the gene was controlled by a tetracyclin promoter, and the recombinant protein produced included an eight amino acid C-terminal tag that allowed affinity chromatography on a *Strep*-Tactin column (IBA) and additional amino acids depending on the construct.

Recombinant protein expression

Isoforms G6PD1, G6PD2, and G6PD6 were produced in *E. coli* strain XL-1 Blue, G6PD3, G6PD4, and G6PD5 in BL21 (DE3) to obtain higher protein yields. The expression strains were routinely cultured overnight at 37°C in LB ampicillin 100 µg ml⁻¹, inoculated at 1/250 dilution in LB ampicillin 100 µg ml⁻¹ and cultured at 28°C. Gene expression was induced by addition of 0.2 mg l⁻¹ LB anhydrotetracyclin at OD₆₀₀ approximately 0.3. At OD₆₀₀ 0.7–0.8, cultures were cooled on ice and cells were harvested by centrifugation at 6000 g. Cells from 250 ml of culture were resuspended in 1 ml of Wash buffer (0.1 M Tris pH8, 1 mM EDTA, 150 mM NaCl) and lysed by sonication. The protein was purified from cell lysate using *Strep*-Tactin columns (IBA) following the manufacturer's instructions.

Fractions with highest protein content were combined and the protein concentration was determined by Bio-Rad DC Protein Assay (Bio-Rad, Hercules, CA, USA).

Enzyme assays and kinetic analyses

All assays were performed in double beam mode using a DW-2RC UV-VIS spectrophotometer (SLM Instruments, Rochester, NY, USA) detecting OD at 340 nm. A sample containing the enzyme was measured against a blank containing 0.1 M Tris (pH 8) with substrates or various DTT and NaTT concentrations as indicated in the text. The increase in absorbance at 340 nm was monitored, which is diagnostic for the generation of NADPH. Saturation curves were obtained by measuring activity under varying concentrations of NADP⁺ and excess amount of G6P, at least 10-fold of saturating NADP⁺. To determine K_{iNADPH} , saturation curves were obtained under three non-saturating concentrations of the inhibitor NADPH. The values of K_m , apparent K_m (K'_m), and V_{max} were deduced from best fitting hyperbolic curves using ORIGIN 5.0 (OriginLab Corporation, Northampton, MA, USA). The K'_m values were plotted against NADPH concentration [I] for each curve. The K_i values were deduced from these plots by extrapolating to $K'_m = 0$, in which $K_i = -[I]$ based on the equation $K'_m = K_m(1 + [I]/K_i)$ (Segel, 1975). Each experiment was repeated two to three times. Buffers of 0.1 M 4-morpholine-ethanesulfonic acid (MES) for pH 6–7.5, 0.1 M Tris for pH 7–9.5, 0.1 M 3-(cyclohexylamino)propanesulfonic acid (CAPS) for pH 9–10 were used for determining the optimal pH under constant substrate concentrations. To study the effect of reduction by DTT, the enzyme was incubated with G6P and 0, 3, 5, 10 mM DTT at time zero. The reaction was initiated by adding constant non-saturating concentrations of NADP⁺ to the mixture and OD₃₄₀ was measured at each time point. To examine its reversibility, 5 mM NaTT was added to reactions containing 3 mM DTT and the reaction was measured in a time-dependent manner. For DTT and NaTT, a 100-mM solution was freshly prepared in water for each experiment.

Semiquantitative PCR for gene expression analysis

Primers were designed to amplify the 3'-UTR of individual isoforms to distinguish the transcripts (Table 4). Semiquantitative PCR was performed using 3% of the RT reaction as template, with 1 U *Taq* polymerase (Roche) in a total volume of 15 µl with 28 cycles (26 for *ACTIN1*). Primer specificity and efficiency were confirmed by amplifying genomic fragments with serial dilution of genomic DNA. All RT-PCRs were repeated two to three times with similar relative levels detected. The *ACTIN1* gene (GenBank accession AY074873) was used as loading control. The relative densities of the bands detected for the transcript of an isoform in various tissues were calculated from the gel image with Molecular Analyst (Bio-Rad), and normalized to the relative intensity calculated for the actin band from the respective tissue. All tissues were harvested from Col-2 plants grown in the same growth chamber for extracting RNA, except for roots, which were harvested from plantlets grown on agar plates.

Protein preparation and zymograms

Protein extraction from various tissues was carried out as previously described (Focks and Benning, 1998). Activity of G6PDH in protein extracts decreased during storage at -80°C. Therefore, all data were obtained with freshly prepared protein extracts. For analyses of activity in different tissues in Figure 4(a), protein samples were concentrated using an ultrafree-0.5 centrifugal filter device with a

nominal MW limit of 50 kDa (Millipore, Bedford, MA, USA) and adjusted to approximately 6.3 mg ml^{-1} . All other experiments were carried out with protein extracts adjusted to $1\text{--}2 \text{ mg ml}^{-1}$ if not stated otherwise. A recombinant G6PDH from *Leuconostoc mesenteroides* (Sigma, St Louis, MO, USA) was used as a positive control. Approximately $0.5\text{--}1 \mu\text{l}$ was applied onto Titan III cellulose acetate (CAE) plates (Helena Laboratories, Beaumont, TX, USA) and electrophoresed for 20 min at 175 mV at 4°C in electrophoresis buffer (180 mM Tris base, 100 mM boric acid, 20 mM dibasic sodiumphosphate, 2 mM EDTA, 10 mM glucose). The CAE plates were overlaid with 5 ml of dissolved (approximately 40°C) reaction medium (0.1 M Tris pH 9.2, 1 mM NADP^+ , 10 mM G6P, 1% agar, 0.02 mg ml^{-1} phenylmethylsulfonate, 0.26 mg ml^{-1} nitroterazolum blue chloride) and incubated at 37°C until the color developed.

T-DNA insertion line isolation

T-DNA insertion lines were obtained from the SALK T-DNA insertion population (Alonso *et al.*, 2003). Genotypes of the plants were determined using PCR primers designed for gene-specific sequences flanking the insertion site and T-DNA left border using the i-sect tool (<http://signal.salk.edu/tdnaprimers.html>) with some modification to increase primer specificity. The homozygosity of the T-DNA insertion was indicated by the presence of a product of one gene-specific primer and the T-DNA left border primer, and the absence of a product of the two gene-specific primers. The insertion site was confirmed by sequencing the PCR product generated with the T-DNA left border primer and the gene-specific primer. To confirm the loss of gene expression, RT-PCR was carried out similarly as described in previous sections except with 600 ng of initial total RNA and 35 cycles of PCR.

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